Algorithms and Data Structures

REPORT LAB 7

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TASK 1

```
rl.zlb' '--pid=Microsoft-MIEngine-Pid-u5lltw4m.wmz' '--dbgExe=C:\TDM-GCC-64\bin\gdb.exe' '
INITIAL RANDOM ARRAY (50)
0.1035  0.8809  0.7368  0.0885  0.3238
0.1016  0.7521  0.0437  0.3554  0.6143
0.4951  0.6612  0.8956  0.0211  0.5780
0.3488  0.4871  0.8950  0.1765  0.3942
0.3218  0.9952  0.5099  0.1500  0.8114
0.1791  0.2293  0.4936  0.3320  0.7929
0.3703  0.5939  0.0494  0.8226  0.0266
0.5158  0.2299  0.5849  0.4005  0.6944
0.9441  0.2418  0.3861  0.8087  0.5693
0.9169  0.3430  0.0875  0.9978  0.6983

DISTRIBUTING ELEMENTS INTO BUCKETS
Element 0.1035 added to Bucket [5]. Current Bucket: 0.1035
Element 0.8809 added to Bucket [44]. Current Bucket: 0.8809
Element 0.7368 added to Bucket [36]. Current Bucket: 0.7368
Element 0.0885 added to Bucket [4]. Current Bucket: 0.0885
Element 0.3238 added to Bucket [16]. Current Bucket: 0.3238
Element 0.1016 added to Bucket [5]. Current Bucket: 0.1016 -> 0.1035
Element 0.7521 added to Bucket [37]. Current Bucket: 0.7521
Element 0.0437 added to Bucket [2]. Current Bucket: 0.0437
Element 0.3554 added to Bucket [17]. Current Bucket: 0.3554
Element 0.6143 added to Bucket [30]. Current Bucket: 0.6143
Element 0.4951 added to Bucket [24]. Current Bucket: 0.4951
Element 0.6612 added to Bucket [33]. Current Bucket: 0.6612
Element 0.8956 added to Bucket [44]. Current Bucket: 0.8809 -> 0.8956
Element 0.0211 added to Bucket [1]. Current Bucket: 0.0211
Element 0.5780 added to Bucket [28]. Current Bucket: 0.5780
Element 0.3488 added to Bucket [17]. Current Bucket: 0.3488 -> 0.3554
Element 0.4871 added to Bucket [24]. Current Bucket: 0.4871 -> 0.4951
Element 0.8950 added to Bucket [44]. Current Bucket: 0.8809 -> 0.8950 -> 0.8956
Element 0.1765 added to Bucket [8]. Current Bucket: 0.1765
Element 0.3942 added to Bucket [19]. Current Bucket: 0.3942
Element 0.3218 added to Bucket [16]. Current Bucket: 0.3218 -> 0.3238
Element 0.9952 added to Bucket [49]. Current Bucket: 0.9952
Element 0.5099 added to Bucket [25]. Current Bucket: 0.5099
Element 0.1500 added to Bucket [7]. Current Bucket: 0.1500
Element 0.8114 added to Bucket [40]. Current Bucket: 0.8114
Element 0.1791 added to Bucket [8]. Current Bucket: 0.1765 -> 0.1791
Element 0.2293 added to Bucket [11]. Current Bucket: 0.2293
Element 0.4936 added to Bucket [24]. Current Bucket: 0.4871 -> 0.4936 -> 0.4951
Element 0.3320 added to Bucket [16]. Current Bucket: 0.3218 -> 0.3238 -> 0.3320
Element 0.7929 added to Bucket [39]. Current Bucket: 0.7929
Element 0.3703 added to Bucket [18]. Current Bucket: 0.3703
Element 0.5939 added to Bucket [29]. Current Bucket: 0.5939
Element 0.0494 added to Bucket [2]. Current Bucket: 0.0437 -> 0.0494
Element 0.8226 added to Bucket [41]. Current Bucket: 0.8226
Element 0.0266 added to Bucket [1]. Current Bucket: 0.0211 -> 0.0266
Element 0.5158 added to Bucket [25]. Current Bucket: 0.5099 -> 0.5158
Element 0.2299 added to Bucket [11]. Current Bucket: 0.2293 -> 0.2299
Element 0.5849 added to Bucket [29]. Current Bucket: 0.5849 -> 0.5939
Element 0.4005 added to Bucket [20]. Current Bucket: 0.4005
Element 0.6944 added to Bucket [34]. Current Bucket: 0.6944
Element 0.9441 added to Bucket [47]. Current Bucket: 0.9441
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Element 0.9441 added to Bucket [47]. Current Bucket: 0.9441
Element 0.2418 added to Bucket [12]. Current Bucket: 0.2418
Element 0.3861 added to Bucket [19]. Current Bucket: 0.3861 -> 0.3942
Element 0.8087 added to Bucket [40]. Current Bucket: 0.8087 -> 0.8114
Element 0.5693 added to Bucket [28]. Current Bucket: 0.5693 -> 0.5780
Element 0.9169 added to Bucket [45]. Current Bucket: 0.9169
Element 0.3430 added to Bucket [17]. Current Bucket: 0.3430 -> 0.3488 -> 0.3554
Element 0.0875 added to Bucket [4]. Current Bucket: 0.0875 -> 0.0885
Element 0.9978 added to Bucket [49]. Current Bucket: 0.9952 -> 0.9978
Element 0.6983 added to Bucket [34]. Current Bucket: 0.6944 -> 0.6983

FINAL SORTED ARRAY
0.0211  0.0266  0.0437  0.0494  0.0875
0.0885  0.1016  0.1035  0.1500  0.1765
0.1791  0.2293  0.2299  0.2418  0.3218
0.3238  0.3320  0.3430  0.3488  0.3554
0.3703  0.3861  0.3942  0.4005  0.4871
0.4936  0.4951  0.5099  0.5158  0.5693
0.5780  0.5849  0.5939  0.6143  0.6612
0.6944  0.6983  0.7368  0.7521  0.7929
0.8087  0.8114  0.8226  0.8809  0.8950
0.8956  0.9169  0.9441  0.9952  0.9978
PS C:\Users\ILGIN>
```

TASK 2

```
CC-64\bin\gdb.exe' '--interpreter=mi'
INITIAL RANDOM ARRAY (100)
78 40 78 16 55 53 63 99 17 77
70 23 64 49 89 81 19 27 18 23
27 89 19 31 36 63 32 79 90 27
91 69 80 88 37 83 21 96 86 26
97 21 50 72 21 56 24 27 56 87
70 23 95 53 40 85 73 40 50 46
94 34 13 72 50 91 66 41 20 22
15 27 34 19 66 21 97 19 16 40
36 47 64 81 20 29 29 84 62 60
59 66 54 20 52 25 13 17 50 82

TRACING RANDOMIZED-SELECT (Finding 20th smallest)
Partitioning subarray [p=0, r=99]: 78 40 78 16 55 53 63 99 17 77 70 23 64 49 89 81 19 27 18 23 27 89 19 31 36 63 32 79 90 27 91 69 80 88 37 83 21 96 86 26 97 21 50 72 21
56 24 27 56 87 70 23 95 53 40 85 73 40 50 46 94 34 13 72 50 91 66 41 20 22 15 27 34 19 66 21 97 19 16 40 36 47 64 81 20 29 29 84 62 60 59 66 54 20 52 25 13 17 50 82
Partitioning subarray [p=0, r=85]: 78 40 78 16 55 53 63 17 77 70 23 64 49 81 19 27 18 23 27 19 31 36 63 32 79 27 69 80 37 83 21 82 26 21 50 72 21 56 24 27 56 70 23 53 40
85 73 40 50 46 34 13 72 50 66 41 20 22 15 27 34 19 66 21 19 16 40 36 47 64 81 20 29 29 84 62 60 59 66 54 20 52 25 13 17 50
Partitioning subarray [p=0, r=51]: 40 16 17 23 49 19 27 18 23 27 19 31 36 32 27 37 21 26 21 50 21 24 27 23 40 40 50 46 34 13 50 41 20 22 15 27 34 19 21 19 16 40 36 47 20
29 29 20 50 25 13 17
Partitioning subarray [p=0, r=29]: 16 17 23 19 27 18 23 27 19 27 21 26 21 21 24 27 23 13 20 22 15 17 19 21 19 16 20 20 25 13
Partitioning subarray [p=15, r=29]: 27 23 23 23 22 27 27 27 21 21 26 21 21 25 24
Partitioning subarray [p=19, r=29]: 22 27 27 27 27 24 26 23 23 25 23
Partitioning subarray [p=19, r=22]: 22 23 23 23
Partitioning subarray [p=19, r=21]: 22 23 23
Partitioning subarray [p=19, r=20]: 22 23
Partitioning subarray [p=19, r=19]: 22

RESULT: The 20th smallest element is: 22
PS C:\Users\ILGIN>
```